

A proof of the conjectured recurrence for OEIS A045742

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Abstract

OEIS A045742 carries a conjectured linear recurrence with polynomial coefficients, of order 3. The entry gives no generating function, but it does give the sequence as a single hypergeometric sum. We prove the conjecture without assuming any generating function: creative telescoping produces, from the summand alone, an operator of order 2 that annihilates the sequence, together with a rational certificate that is checked as an exact identity rather than trusted. The conjectured operator is then shown to be a left multiple of it by division in the Ore algebra $\mathbb{Q}(n)[N]$. That last step is what the conjecture needs: the conjectured recurrence has order 3 while the summand only needs order 2, so it is not itself a telescoper and cannot be produced directly.

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1 The sequence and the conjecture

OEIS A045742 is “Number of interior faces in all noncrossing connected graphs on n nodes on a circle”. It has offset 2 and begins

$$a(2), \dots, a(9) = 0, 1, 13, 141, 1456, 14778, 149031, 1499773, \dots$$

The entry states, not as a conjecture,

$$a(n) = \text{Sum}_{k=1..n-2} k \cdot \text{binomial}(n+k-2, k) \cdot \text{binomial}(3n-3, n-2-k) / (n-1).$$

so that, writing $F(n, k)$ for the summand,

$$a(n) = \sum_{k=1}^{n-2} F(n, k), \quad F(n, k) = \frac{k \binom{3n-3}{-k+n-2} \binom{k+n-2}{k}}{n-1}. \quad (1)$$

This is the only input the proof takes from the entry, and it was checked against the entry’s own DATA before use.

The entry separately carries the following comment:

Conjecture D-finite with recurrence $n(n-1)(523n-1993)a(n) - 6(n-1)(759n^2-6019n+12790)a(n-1) - 12(n-2)(4707n^2-17937n+6260)a(n-2) + 72(3n-10)(759n-2224)(3n-11)a(n-3) = 0$.
- R. J. Mathar, Jul 26 2022

As of the “Last modified” line on the live entry (2022-07-26, revision 20) the statement is still recorded as a conjecture, and no proof appears among the entry’s links, formulas or programs.

2 Telescoping the summand

Let N denote the shift in n , $(Nf)(n) = f(n+1)$, and Δ_k the forward difference in k , $(\Delta_k g)(k) = g(k+1) - g(k)$.

Lemma 1. *Suppose there are rational functions $\sigma_0, \dots, \sigma_r$ of n , not all zero, and a rational function $R(n, k)$ such that*

$$\sum_{i=0}^r \sigma_i(n) F(n+i, k) = \Delta_k [R(n, k) F(n, k)]. \quad (2)$$

Suppose further that $F(n, k) = 0$ for every k outside the range of summation in (1). Then $\sum_i \sigma_i(n) a(n+i) = 0$.

Proof. Because $F(n, k)$ vanishes off the stated range, the sum in (1) may be taken over all $k \in \mathbb{Z}$ without changing it, and the same holds for each $F(n+i, k)$. Summing (2) over all k , the left side becomes $\sum_i \sigma_i(n) a(n+i)$ and the right side telescopes: the partial sums of a forward difference collapse to the boundary values, which are 0 because $R(n, k) F(n, k)$ vanishes for $|k|$ large. Hence the left side is 0. $\square$

The point of Lemma 1 is that (2) is an identity between rational functions once both sides are divided by $F(n, k)$: the quotients $F(n+i, k)/F(n, k)$ and $F(n, k+1)/F(n, k)$ are rational because F is hypergeometric in both variables. So a proposed (σ, R) can be *checked*, exactly, whatever produced it.

For A045742 such a pair exists with $r = 2$:

$$\sigma_0(n) = -2, \quad \sigma_1(n) = \frac{-9n^3 - 33n^2 + 8n + 32}{54n^5 + 9n^4 - 114n^3 + 77n^2 - 14n}, \quad \sigma_2(n) = \frac{6n^5 + 7n^4 - 24n^3 - 23n^2 + 18n + 10}{324n^5 + 54n^4 - 684n^3 + 462n^2 - 84n}$$

and certificate

$$R(n, k) = \frac{12k^5n^3 + 14k^5n^2 - 14k^5n + 60k^4n^4 + 112k^4n^3 - 21k^4n^2 - 49k^4n + 54k^3n^5 + 207k^3n^4 + 126k^3n^3 - 12k^3n^2 - 12k^3n + 12k^2n^5 + 12k^2n^4 - 12k^2n^3 - 12k^2n^2 - 12k^2n + 12k^1n^5 + 12k^1n^4 - 12k^1n^3 - 12k^1n^2 - 12k^1n + 12k^0n^5 + 12k^0n^4 - 12k^0n^3 - 12k^0n^2 - 12k^0n + 12k^0}{12k^5n^3 + 14k^5n^2 - 14k^5n + 48k^4n^4 + 104k^4n^3 - 56k^4n^2 + 12k^3n^5 + 14k^3n^4 - 12k^3n^3 - 12k^3n^2 - 12k^3n + 12k^2n^5 + 12k^2n^4 - 12k^2n^3 - 12k^2n^2 - 12k^2n + 12k^1n^5 + 12k^1n^4 - 12k^1n^3 - 12k^1n^2 - 12k^1n + 12k^0n^5 + 12k^0n^4 - 12k^0n^3 - 12k^0n^2 - 12k^0n + 12k^0}$$

Dividing (2) by $F(n, k)$ and cancelling gives 0 identically, so Lemma 1 applies and

$$L = \sum_{i=0}^2 \sigma_i(n) N^i \quad (3)$$

annihilates a .

3 From the derived recurrence to the conjectured one

The operator (3) is not the conjectured one. It has order 2, below the conjectured order 3: the conjecture is a consequence of a shorter relation, not the shortest relation itself. The two are connected by division in the Ore algebra $\mathcal{A} = \mathbb{Q}(n)[N]$, where multiplication obeys $Nq(n) = q(n+1)N$.

Lemma 2. *If $C = QL$ in $\mathcal{A}$ and $L(a) = 0$, then $C(a) = 0$ at every n where the coefficients of Q are defined.*

Proof. $C(a) = (QL)(a) = Q(L(a)) = Q(0) = 0$. The coefficients of Q are rational functions of n , so the step is valid away from their poles. $\square$

Writing the conjectured recurrence as an operator C and dividing by L leaves remainder 0, with quotient

$$Q = (-245916n^3 + 228744n^2 - 37476n - 3816) + \left(\frac{169452n^6 + 738126n^5 + 739980n^4 - 142926n^3 - 510624n^2}{6n^3 + 7n^2 - 7n} \right)$$

The coefficients of Q are rational functions of n ; their only integer poles are $n \in \{0\}$, below the range claimed here.

Theorem 3. *The conjectured recurrence*

$$n(n-1)(523n-1993)a(n)-6(n-1)(759n^2-6019n+12790)a(n-1)-12(n-2)(4707n^2-17937n+6260)a(n-2)$$

holds for every $n \geq 5$.

Proof. By Lemma 1 the operator (3) annihilates a , and by Lemma 2 applied to $C = QL$ so does the conjectured operator. The integer poles of Q all lie below $n = 5$. $\square$

4 Verification

Every number above is an output of a script written separately from this note, and four independent checks were run.

First, the certificate was verified, not trusted: (2) was divided by $F(n, k)$ and the difference of the two sides cancelled to 0 as a rational function of n and k .

Second, the vanishing of $F(n, k)$ off the range of summation — the hypothesis Lemma 1 needs, and the one that is easy to assume without looking — was checked directly, on concrete integers on both sides of the range.

Third, the factorisation was confirmed by multiplying back: QL was expanded in $\mathcal{A}$ and its coefficients cancelled against those of C to zero.

Fourth, the conjecture itself was evaluated on the published terms, in exact integer arithmetic and without reference to any operator: for each n in range the sum $\sum_i p_i(n)a(n-i)$ was formed from the entry's own values and found to vanish, for all 16 values of n from $n = 5$ upward. The formula (1) was likewise checked term by term against the DATA section before being used.

References

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